Supplementary Material: Sharp finite-size coexistence in an adaptive multi-species competitive network: scaling evidence for a continuous transition

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S1. Binder data-collapse details

If the transition is asymptotically continuous, the Binder curves should collapse onto a single master curve under the standard finite-size scaling form

$$U_L(\mu) = \tilde{U}\big((\mu - \mu_c)\, L^{1/\nu}\big)\,, \tag{S1}$$

with two free parameters μ_c and $1/\nu$. We attempt this collapse by minimising the per-point inter-curve standard deviation of U_L on a common grid of the scaling variable $(\mu - \mu_c)L^{1/\nu}$, restricted to the dip region $|\mu - \mu_c| \leq 0.012$ (outside this window all curves trivially asymptote to $2/3$). $L = 64$ is excluded from the fit: its anomalously deep negative dip on a system of only 4096 nodes is a small-system fluctuation effect that no two-parameter rescaling can absorb.

The three-size fit $L \in \{128, 256, 384\}$ does *not* collapse cleanly (Fig. S1, middle panel): the $L = 128$ dip is much deeper than any rescaling allows, leaving residual scatter $\sigma_U \approx 0.04$. Restricting to the two largest sizes $L \in \{256, 384\}$ tightens the collapse by an order of magnitude, $\sigma_U \approx 0.005$. Bootstrap 95% confidence intervals from 200 seed-resamplings give $\mu_c \in [0.346, 0.348]$ and $1/\nu \in [0.97, 1.99]$. The point estimate $1/\nu \approx 1.55$ should not be over-interpreted: two curves fitted with two free parameters provide a weakly constrained fit, and the $1/\nu$ confidence interval spans most plausible two-dimensional continuous-transition exponents. The collapse therefore confirms only that $L = 256$ and $L = 384$ are mutually consistent with a standard FSS form, not a specific universality class.

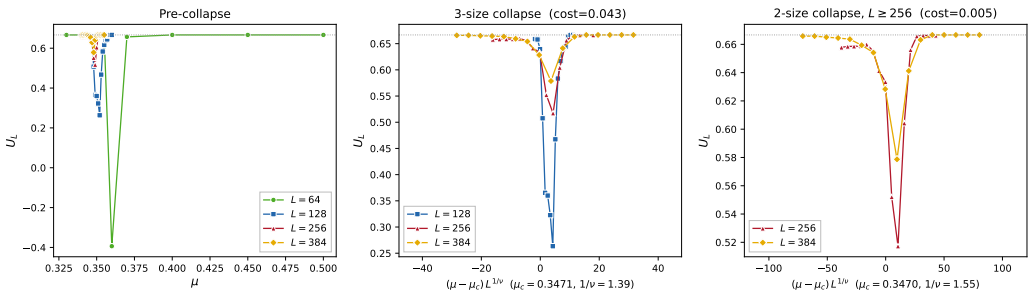

Figure S1: **Binder data-collapse attempt. Left:** raw U_L vs. μ for all four sizes. **Middle:** collapse fit using $L \in \{128, 256, 384\}$; the $L = 128$ dip is much deeper than any two-parameter rescaling can absorb (mean inter-curve std $\sigma_U \approx 0.04$). **Right:** restricting to the two largest sizes $L \in \{256, 384\}$ tightens the collapse to $\sigma_U \approx 0.005$. The contrast is consistent with $L = 128$ lying outside the regime where the leading-order finite-size scaling form holds.

The contrast between the failed three-size and acceptable two-size collapses is consistent with the standard corrections-to-scaling picture of an approach to a continuous transition: $L = 128$ lies outside the regime where the leading-order finite-size scaling form holds; $L \geq 256$ is more consistent with it. Definitively settling whether the implied universality class is robust would require either simulations at $L \geq 512$ or systematic correction-to-scaling fits with additional free parameters.

S2. Per-seed ρ_b distributions at $L = 384$

Figure S2 shows the per-seed ρ_b distribution at $L = 384$ across the Binder minimum, analogous to Figure 4 of the main text at $L = 128$. At $\mu = 0.345$ the distribution is a tight unimodal peak in the ordered phase ($\rho_b \approx 0.20$); at $\mu = 0.351$ all 64 seeds have collapsed to a tight unimodal peak in the disordered phase ($\rho_b \approx 0.55$). Bimodality persists at the largest size, but only inside a narrow window $\mu \in [0.347, 0.349]$ around the Binder minimum—substantially narrower than the $L = 128$ bimodal window. The narrowing of the bimodal window with L tracks the same sharpening-yet-continuous trend as the Binder shallowing reported in the main text.

S3. Convergence diagnostics

Figure S3 shows ρ_b time-series for 8 seeds at three μ values at $L = 256$ (snapshots every 5 000 frames, 100 000 total). Away from the transition ($\mu =$

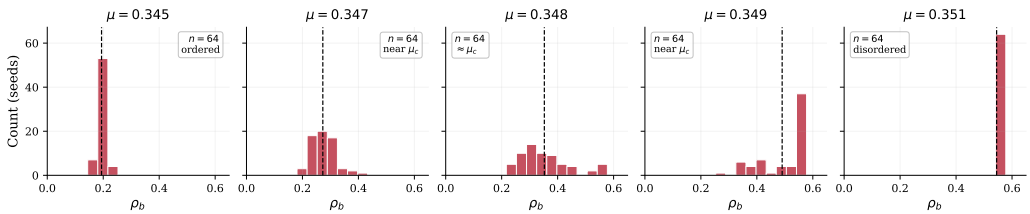

Figure S2: **Distribution of per-seed ρ_b at $L = 384$ across the Binder minimum** (64 seeds per panel, 200 000 frames). Bimodality persists at the largest system size but is confined to a narrow window $\mu \in [0.347, 0.349]$ around the Binder minimum ($\mu_c \approx 0.348$). Vertical dashed lines mark the mean.

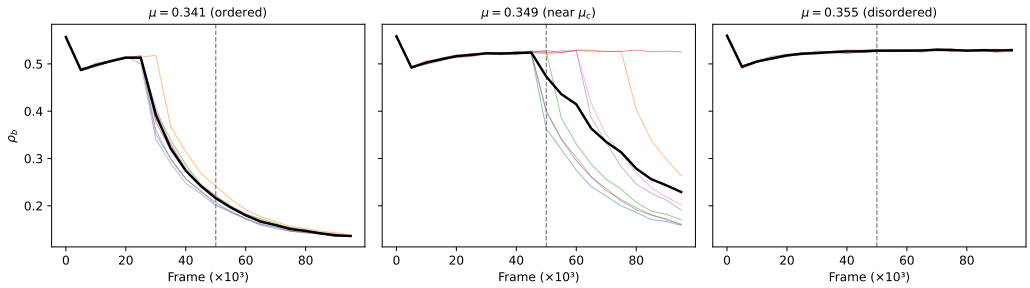

Figure S3: **Convergence diagnostics at $L = 256$.** ρ_b vs. frame for 8 seeds at three μ values (snapshots every 5 000 frames, 100 000 total). Dashed line marks the burn-in cutoff (second-half averages are reported in the main text). At $\mu = 0.341$ (ordered) and 0.355 (disordered), seeds converge within the burn-in window. At $\mu = 0.349$ (near μ_c), some seeds switch basins during the second half.

0.341 and 0.355), second-half averages are stable. Near μ_c ($\mu = 0.349$), some seeds undergo within-run phase switching, so reported second-half averages for those seeds conflate the two basins. Spot-checks at $L = 384$ (200 000 frames) show similar within-run basin-switching near μ_c; the doubled run length improves but does not eliminate this near-critical residence-mixing effect.